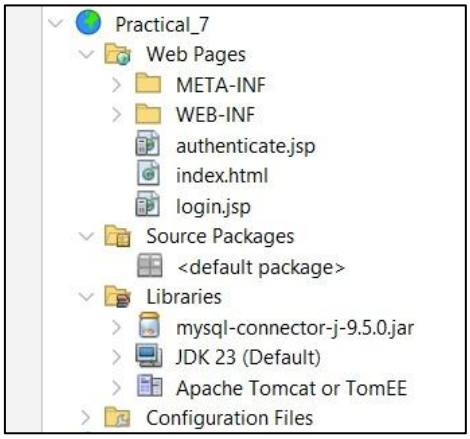
PRACTICAL 7

JSP - User Authentication Develop a data driven JSP application for user authentication.



CODE:

Login.jsp

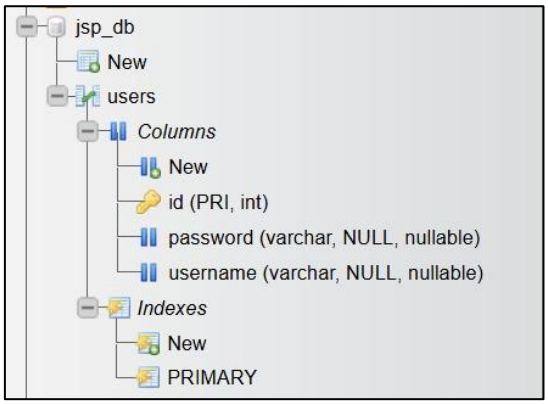
```
<%@ page language="java" %>
<html>
<head>
   <title>User Login</title>
</head>
<body>
<h2>User Authentication - JSP</h2>
<form action="authenticate.jsp" method="post">
   Username: <input type="text" name="username"><br><br>
   Password: <input type="password" name="password"><br><br>
   <input type="submit" value="Login">
</form>
</body>
</html>
```

authenticate.jsp

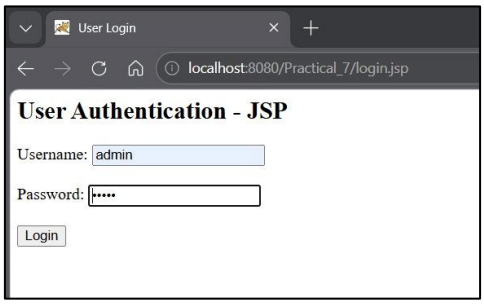
```
<%@ page import="java.sql.*" %>
<%
    String user = request.getParameter("username");
    String pass = request.getParameter("password");
    boolean valid = false;
    try {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3307/jsp_db",
            "root", ""
        )
        PreparedStatement ps = con.prepareStatement(
            "SELECT * FROM users WHERE username=? AND password=?"
        );
        ps.setString(1, user);
        ps.setString(2, pass);
        ResultSet rs = ps.executeQuery();
        if (rs.next()) {
            valid = true;
        }
        rs.close();
        ps.close();
        con.close();
    } catch (Exception e) {
        out.println("Error: " + e);
    }
    if (valid) {
%>
```

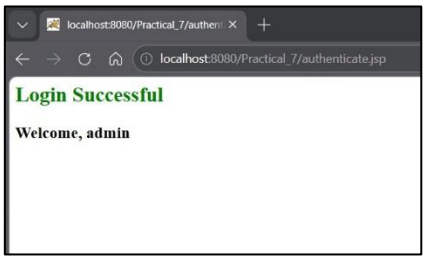
```
    <h2 style="color:green">Login Successful</h2>
    <h3>Welcome, <%= user %></h3>
<%
  } else {
%>
    <h2 style="color:red">Invalid Username or Password</h2>
<%
  }
%>
```

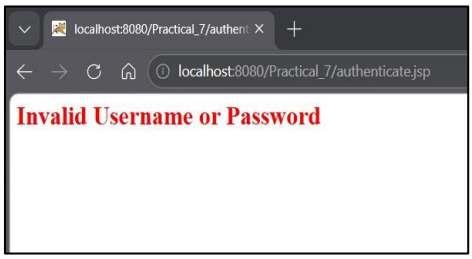
Database



OUTPUT:

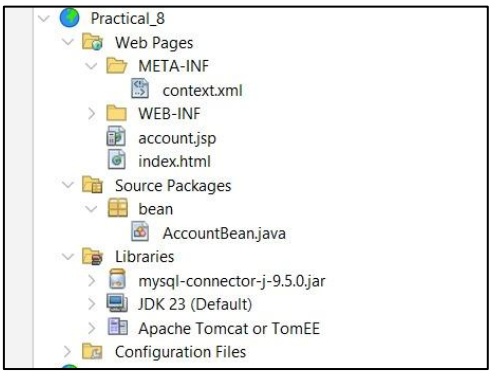






PRACTICAL 8

JSP - Java Beans Develop an application for Bank Account information using Java Beans. This application will perform various operation on bank account like withdraw, deposit and displaying the information.



CODE

AccountBean.java

```
package bean

import java.io.Serializable;

public class AccountBean implements Serializable {

    private int accountNo;

    private String name;

    private double balance;

    // No-argument constructor

    public AccountBean() {

        balance = 0.0;

    }

    // Getters and Setters

    public int getAccountNo() {

        return accountNo;

    }

    public void setAccountNo(int accountNo) {

        this.accountNo = accountNo;

    }
```

```
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public double getBalance() {
        return balance;
    }
    // Business methods
    public void deposit(double amount) {
        balance += amount;
    }
    public boolean withdraw(double amount) {
        if (amount <= balance) {
            balance -= amount;
            return true;
        }
        return false;
    }
}
```

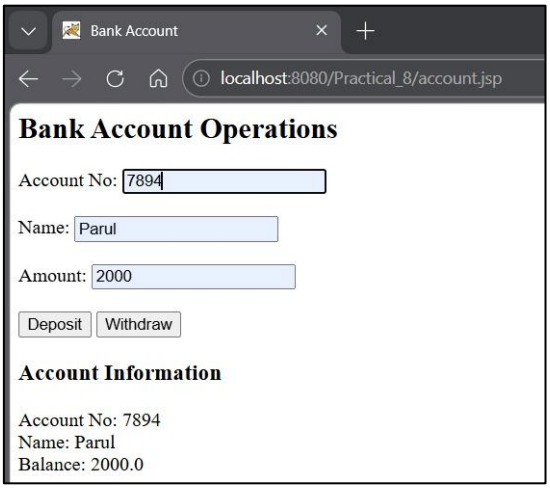
account.jsp

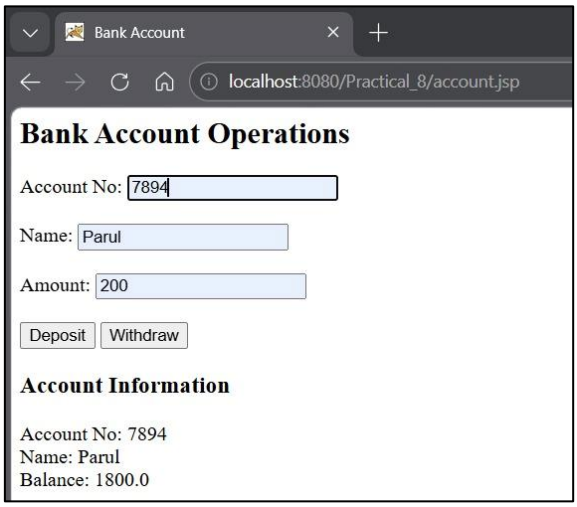
```
<%@ page import="bean.AccountBean" %>
<jsp:useBean id="acc" class="bean.AccountBean" scope="session" />

<html>
<head>
    <title>Bank Account</title>
</head>
```

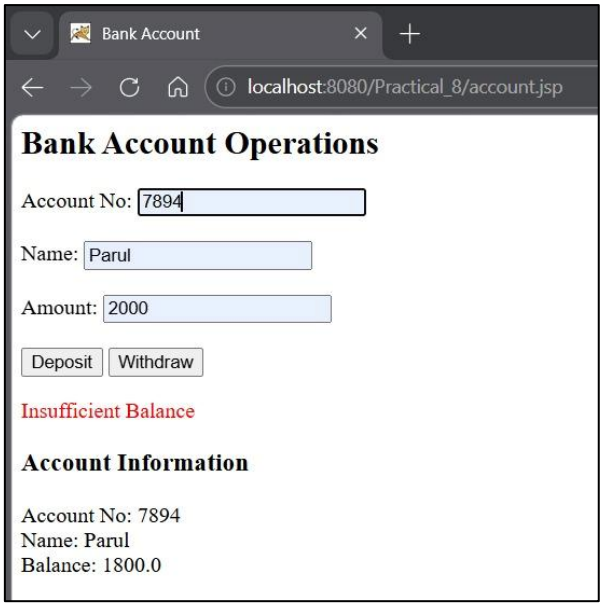
```
<body
<h2>Bank Account Operations</h2>
<form method="post">
   Account No: <input type="text" name="ano"><br><br>
   Name: <input type="text" name="name"><br><br>
   Amount: <input type="text" name="amount"><br><br>
   <input type="submit" name="action" value="Deposit">
   <input type="submit" name="action" value="Withdraw">
</form>
<%
   String action = request.getParameter("action");
   if (action != null) {
      int ano = Integer.parseInt(request.getParameter("ano"));
      String name = request.getParameter("name");
      double amount = Double.parseDouble(request.getParameter("amount"));
      acc.setAccountNo(ano);
      acc.setName(name);
      if (action.equals("Deposit")) {
         acc.deposit(amount);
      } else if (action.equals("Withdraw")) {
         boolean success = acc.withdraw(amount);
         if (!success) {
            out.println("<p style='color:red'>Insufficient Balance</p>");
         }
      }
   }
%>
<h3>Account Information</h3>
Account No: <%= acc.getAccountNo() %><br>
```

```
Name: <%= acc.getName() %><br>

Balance: <%= acc.getBalance() %><br>

</body>

</html>
```

OUTPUT

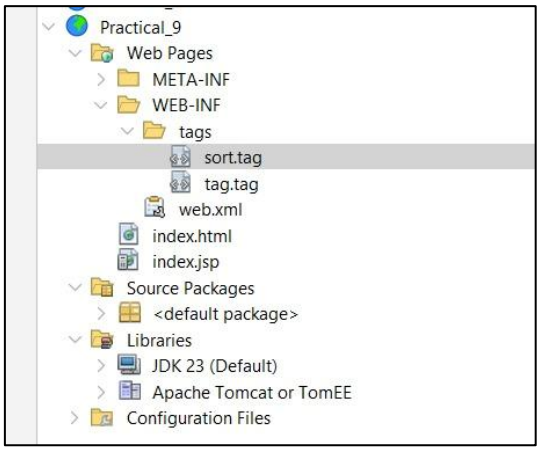






PRACTICAL 9

JSP - Custom Tags Develop a custom JSP tag which accepts 10 numbers from user and sorts them in specified order.



CODE:

Sort.tag

```
<%@ tag body-content="empty" %>

<%@ attribute name="numbers" required="true" %>

<%

    String input = numbers;

    String[] arr = input.split(",");
```

```
    int[] nums = new int[arr.length];
    for(int i=0;i<arr.length;i++){
        nums[i] = Integer.parseInt(arr[i].trim());
    }
    // Sorting (Ascending)
    for(int i=0;i<nums.length;i++){
        for(int j=i+1;j<nums.length;j++){
            if(nums[i] > nums[j]){
                int temp = nums[i];
                nums[i] = nums[j];
                nums[j] = temp;
            }
        }
    }
%>
<h3>Sorted Numbers:</h3>
<%
    for(int n : nums){
        out.print(n + " ");
    }
%>
```

tag.tag

```
<%--
    Document   : tag
    Created on : 18 Feb 2026, 11:53:44 pm
    Author     : DELL
--%>
<%@tag description="put the tag description here" pageEncoding="UTF-8"%>
```

```
<%-- The list of normal or fragment attributes can be specified here: --%>
<%@attribute name="message"%>
<%-- any content can be specified here e.g.: --%>
<h2>${message}</h2>
```

Web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
        http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd"
        version="3.1">

    <welcome-file-list>
        <welcome-file>index.jsp</welcome-file>
    </welcome-file-list>
</web-app>
```

Index.jsp

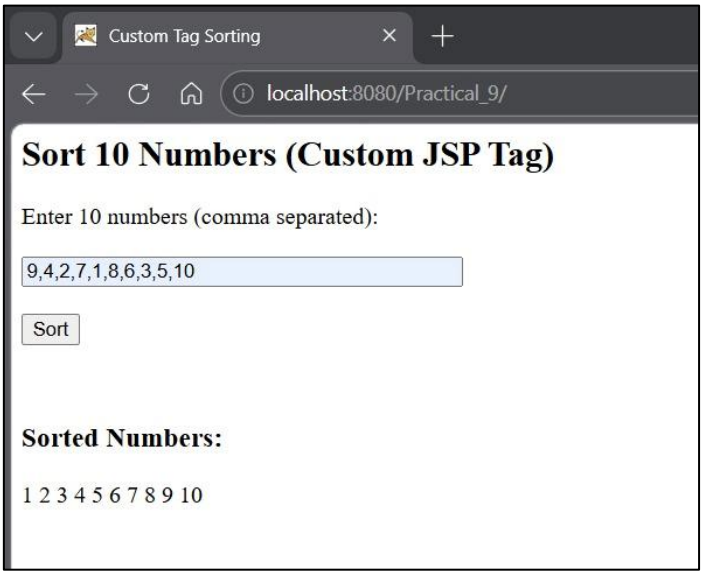
```
<%@ page contentType="text/html" pageEncoding="UTF-8" %>
<%@ taglib prefix="st" tagdir="/WEB-INF/tags" %>
<html>
<head>
    <title>Custom Tag Sorting</title>
</head>
<body>
<h2>Sort 10 Numbers (Custom JSP Tag)</h2>
<form method="post">
    Enter 10 numbers (comma separated): <br><br>
    <input type="text" name="nums" size="40" required>
    <br><br>
    <input type="submit" value="Sort">
</form>
```

```
<br>
<%
   String numbers = request.getParameter("nums");
   if(numbers != null){
%>
<st:sort numbers="<%=numbers%>" />
<%
   }
%>
</body>
</html>
```

SortTag

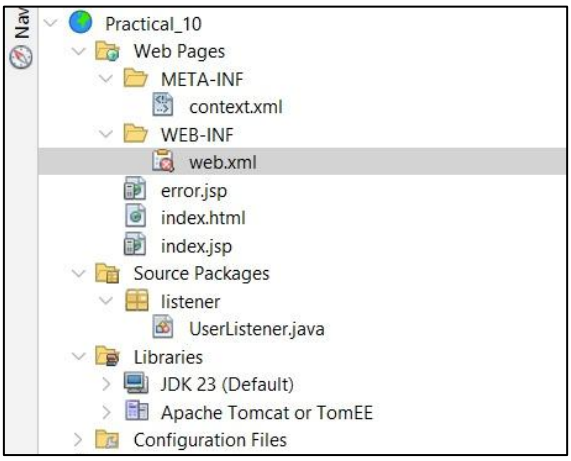
```
public class SortTag {
}
```

OUTPUT:



PRACTICAL 10

JSP - Application Context Develop an application that will allow maximum 3 users to access the application. If number of users exceeds than limit then user will be redirected to the error page.



CODE:

error.jsp

```
<html>
<head>
   <title>Limit Exceeded</title>
</head>
<body
<h1 style="color:red;">
Maximum 3 Users Allowed!
Please try again later.
</h1>
</body>
</html>
```

Index.jsp

```
<%@ page contentType="text/html" pageEncoding="UTF-8"%>
<%
   Integer users = (Integer)application.getAttribute("users");
```

```
    if(users > 3){
        response.sendRedirect("error.jsp");
        return;
    }
%>
<html>
<head>
    <title>Home Page</title>
</head>
<body
<h2>Welcome User!</h2>
<h3>Active Users: <%= users %></h3>
</body>
</html>
```

UserListener.java

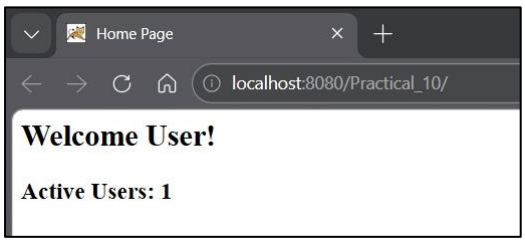
```
package listener;
import javax.servlet.ServletContext;
import javax.servlet.http.HttpSessionEvent;
import javax.servlet.http.HttpSessionListener;
public class UserListener implements HttpSessionListener
    public void sessionCreated(HttpSessionEvent se) {
        ServletContext context = se.getSession().getServletContext();
        Integer userCount = (Integer) context.getAttribute("users");
        if(userCount == null)
            userCount = 0;
        userCount++
        context.setAttribute("users", userCount);
        System.out.println("User Entered: " + userCount);
    }
```

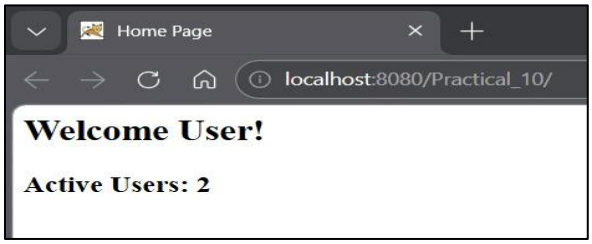
```
    public void sessionDestroyed(HttpSessionEvent se) {
        ServletContext context = se.getSession().getServletContext();
        Integer userCount = (Integer) context.getAttribute("users");
        if(userCount == null)
            userCount = 0;
        userCount--;

        context.setAttribute("users", userCount);
        System.out.println("User Left: " + userCount);
    }
}
```

Web.xml

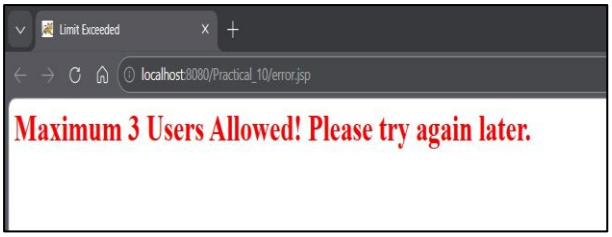
```
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
         version="3.1">
    <listener>
        <listener-class>listener.UserListener</listener-class>
    </listener>
    <welcome-file-list>
        <welcome-file>index.jsp</welcome-file>
    </welcome-file-list>
</web-app>
```

OUTPUT:





Home Page
localhost:8080/Practical_10/
Welcome User!
Active Users: 3


Limit Exceeded
localhost:8080/Practical_10/error.jsp
Maximum 3 Users Allowed! Please try again later.